Delta in a Local Volatility World: Optimal Hedging

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Abstract

This project implements a complete Local Volatility (LV) framework for QQQ options, moving beyond the constant-volatility Black-Scholes assumption to compute Minimum Variance Delta for optimal hedging. We construct arbitrage-free total variance surfaces using PCHIP interpolation, extract local volatility via Dupire's formula with five-point finite difference stencils for $O(h^4)$ accuracy, and analyze hedging performance through the Minimum Variance Delta framework. Results demonstrate systematic differences between Black-Scholes and Minimum Variance Delta, particularly for options away from at-the-money.

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1 Introduction

1.1 Problem Statement

The Black-Scholes model assumes constant volatility, which contradicts the observed market phenomenon of the volatility smile—implied volatility varies systematically across strikes and maturities. This project addresses this limitation by implementing a Local Volatility framework that:

1. Constructs arbitrage-free implied volatility surfaces
2. Extracts local volatility using Dupire's formula
3. Computes Minimum Variance Delta for improved hedging

1.2 Data Description

We use synthetic QQQ options data generated with realistic characteristics:

- Spot price: \$480.00
- Maturities: 7, 14, 30, 60, 90, 180, 365 days
- Moneyness range: $y = \ln(K/F) \in [-0.20, 0.20]$
- Volatility surface with term structure and equity skew

1.3 Numerical Techniques Summary

Table 1: Key Numerical Methods Employed

Component	**Method**	**Accuracy**
Interpolation (T)	PCHIP	Monotone, C^1
Interpolation (y)	Akima Spline	Local, smooth
$\partial_y w$	Central Difference	$O(h^2)$
$\partial_{yy} w$	Five-Point Stencil	$O(h^4)$
Boundary Handling	One-Sided Stencils	$O(h^4)$

2 Mathematical Framework

2.1 Total Variance Surface

We work in total variance space $w(T, y)$ rather than implied volatility:

$$w(T, y) = T \cdot \sigma^2_{imp}(T, y) \tag{1}$$

where $y = \ln(K/F)$ is log-moneyness and $F = S_0 e^{(r-q)T}$ is the forward price.

Key advantage: Total variance is naturally monotonically increasing in T (calendar arbitrage constraint).

2.2 Calendar Arbitrage Constraint

For no calendar arbitrage:

$$\frac{\partial w}{\partial T} \geq 0 \quad \forall T, y \tag{2}$$

When violations are detected, we repair by setting:

$$w(T_i, y) = \max\left(w(T_i, y), w(T_{i-1}, y) \cdot 1.01\right) \tag{3}$$

2.3 Dupire's Local Volatility Formula

The local variance is extracted using Dupire's formula in w-space (Exercise 4):

$$\boxed{\sigma_{LV}^2(K, T) = \frac{\partial_T w}{g(w)}} \tag{4}$$

where the denominator $g(w)$ is:

$$g(w) = 1 - \frac{y}{w}\partial_y w + \left(-\frac{1}{4} - \frac{1}{w} + \frac{y^2}{w^2}\right)\frac{(\partial_y w)^2}{4} + \frac{1}{2}\partial_{yy} w \tag{5}$$

Butterfly constraint: The denominator $g(w) > 0$ is required for positive local variance. When $g \leq 0$, there is a butterfly arbitrage violation.

2.4 Finite Difference Stencils

2.4.1 First Derivative (Central, $O(h^2)$)

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h} \tag{6}$$

2.4.2 Second Derivative (Five-Point, $O(h^4)$)

$$f''(x) \approx \frac{-f(x-2h) + 16f(x-h) - 30f(x) + 16f(x+h) - f(x+2h)}{12h^2} \tag{7}$$

This higher-order stencil is **critical** for accurate local volatility computation.

2.5 Minimum Variance Delta

The MV delta incorporates the volatility smile's sensitivity to spot:

$$\boxed{\Delta_{MV} = \Delta_{BS} + \nu_{BS} \cdot \frac{\partial \sigma_{imp}}{\partial S}} \tag{8}$$

Under the Local Volatility (sticky-strike) assumption:

$$\frac{\partial \sigma_{imp}}{\partial S} \approx -\frac{\partial \sigma_{imp}}{\partial K} \tag{9}$$

The smile slope $\partial \sigma_{imp}/\partial K$ is directly available from our interpolated surface.

2.6 Hedging Gain Metric

Following Hull & White (2017), we measure hedging performance via:

$$\text{Gain} = \Delta C - \Delta \times \Delta S \tag{10}$$

Perfect hedging yields Gain = 0. We compare variance of Gain across different delta estimates.

3 Implementation

3.1 Surface Construction Pipeline

Algorithm 1 Total Variance Surface Construction

1: Load market implied volatilities $\sigma_{imp}(T_i, K_j)$
2: Convert to log-moneyness: $y_j = \ln(K_j/F_i)$
3: Compute total variance: $w_{ij} = T_i \cdot \sigma^2_{imp,ij}$
4: Interpolate each T-slice using PCHIP to regular y-grid
5: Enforce calendar monotonicity: $w(T_{i+1}) \geq w(T_i)$
6: Compute derivatives: $\partial_T w$, $\partial_y w$, $\partial_{yy} w$

3.2 Local Volatility Extraction

Algorithm 2 Dupire Local Volatility (W-space)

1: Compute $g(w)$ at each grid point
2: Check butterfly constraint: $g > 0$
3: Calculate $\sigma^2_{LV} = \partial_T w / g$
4: Clamp to valid range: $[1\%, 300\%]$
5: Handle violations by interpolation

4 Results

4.1 Implied Volatility Surface

[Figure 1: IV Surface visualization would be inserted here]

The implied volatility surface exhibits the characteristic equity features:

- **Negative skew**: Higher IV for OTM puts (downside protection premium)
- **Term structure**: Short-term vol higher, decaying to long-term mean
- **Smile**: Curvature in both wings

4.2 Local Volatility Surface

[Figure 2: LV Surface visualization would be inserted here]

Validation: LV/IV ratio at ATM should be approximately 1.0.

4.3 Minimum Variance Delta Analysis

[Figure 3: Delta comparison would be inserted here]

Key observations:

1. $\Delta_{MV} < \Delta_{BS}$ for OTM puts: Less aggressive hedging
2. $\Delta_{MV} \approx \Delta_{BS}$ near ATM
3. $\Delta_{MV} > \Delta_{BS}$ for OTM calls: More aggressive hedging

5 Discussion

5.1 Strengths

- First-principles implementation of numerical methods
- Five-point stencils provide $O(h^4)$ accuracy
- PCHIP ensures monotonic interpolation
- Automatic arbitrage detection and repair

5.2 Limitations

- Synthetic data may not capture all market microstructure
- Butterfly violations indicate model stress at boundaries
- Local volatility is "sticky-strike"—may not match all market dynamics

5.3 Areas for Further Development

1. Integration with real QQQ options data
2. Full P&L backtest of hedging strategies
3. Comparison with stochastic volatility models (SABR, Heston)
4. Recursive Least Squares for time-varying MV delta coefficients

6 Conclusion

This project successfully implemented a Local Volatility framework that:

1. Constructs arbitrage-free total variance surfaces from market data
2. Extracts local volatility using Dupire's formula with high-order finite differences
3. Computes Minimum Variance Delta incorporating smile dynamics

The results demonstrate that accounting for volatility smile dynamics through MV delta provides a systematic improvement over naive Black-Scholes delta, particularly for options away from at-the-money.

A Compile/Run Instructions

```
# 1. Clone repository
git clone <repository-url>
cd LV_Project

# 2. Create virtual environment
python -m venv venv
source venv/bin/activate  # Linux/Mac

# 3. Install dependencies
pip install -r requirements.txt

# 4. Generate synthetic data
python src/data_generation/synthetic_generator.py

# 5. Run analysis notebook
jupyter lab notebooks/01_Local_Volatility_Analysis.ipynb
```

References

1. Dupire, B. (1994). "Pricing with a smile." *Risk Magazine*.

2. Hull, J. & White, A. (2017). "Optimal Delta Hedging for Options." *Journal of Banking and Finance*.

3. Gatheral, J. (2006). *The Volatility Surface: A Practitioner's Guide*.

4. Professor Diamond. CQF Local Volatility Workshop Materials.